

TCS Agent for KMTNet system

1. Introduction to TCS Agent

TCS Agent is a program included in the Ohio State Instrument Control and Image Acquisition System (ICIMACS). TCS Agent is an interactive program used to monitor and control a telescope control system (TCS). TCS Agent has the roles and functions as follows:

- A software interface between the camera system and the telescope control system
- A translator of ICIMACS Messaging Protocol (IMP) and the low-level protocol of TCS
- A user command-line interface to control and monitor the TCS
- TCS status and telemetry data acquisition and management
- Processing the high-level commands for telescope operation and monitoring
- Monitoring and diagnosing statuses of the telescope, TCS and the communication links

TCS Agent provides a basic IMPv2-compliant interface to TCS. The IMPv2-compliant interface is a remote-command interface for the ICIMACS and End-User software. TCS Agent also provides a command-line interface for the local user.

TCS Agent records and translates the TCS telemetry data into an IMPv2-compliant status message, as well as providing some low-level diagnostics. When the user or ISIS queries the telemetry data with the '~status' command, TCS Agent return the telemetry data and statuses with a message string according to the IMPv2 format.

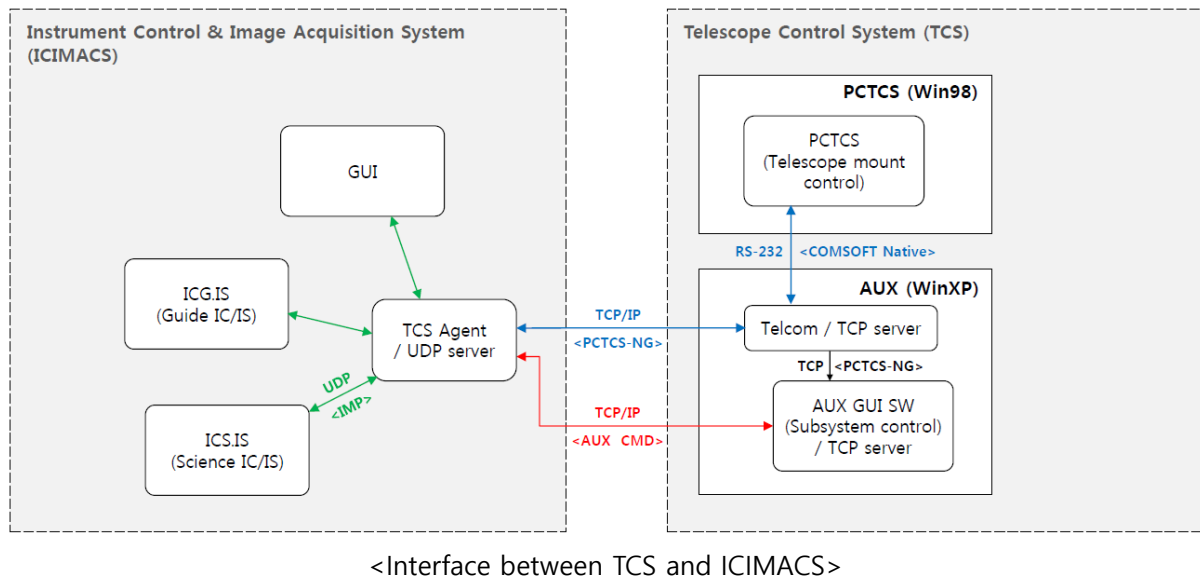
TCS Agent provides high-level commands for telescope operation such as guiding offset, goto RA/Dec, change filters and adjust focus/tip-tilt. The high-level commands are processed using low-level data and remote commands of TCS.

TCS Agent monitors and diagnoses statuses of the telescope, TCS and the communication links. If judged to be an error or abnormal status, TCS Agent prints an error or warning message on the console for the local user, and also provides all statuses and diagnostic data to the remote user with the '~status' command. 'ERROR' message is printed in red color if a critical error was occurred or an unavailable command is received. 'Warning' message is printed in cyan color, if unhealthy situation or an error was occurred but it is not critical.

2. TCS Agent for KMTNet system

2.1. KMTNet TCS

KMTNet TCS has both TCP/IP network interfaces for remote interaction with the ComSoft PC-TCS and the AUX control software. The PC-TCS controls the telescope mount and the AUX control software controls several subsystems of the telescope such as filter/shutter, focuser, dome shutter, and mirror cooling system. The network interface for PC-TCS is implemented with a program named Telcom. Telcom includes a TCP server to process the remote commands according to the "PCTCS-NG Network Protocol". The network interface for AUX control is also implemented with a TCP server in the AUX control software. AUX server processes the remote commands according to the "KMTNet Auxiliary control software - Remote commands definition". TCA Agent acts as a TCP/IP client of both Telcom and AUX server and as an UDP client in ICIMACS and plays a role as a bridge between the telescope and the camera system.



2.2. Modification for KMTNet system

TCS Agent was modified for the KMTNet system. The original version is PCTCS Agent v3.3.1, being used for the Yale 1-m telescope and ANDYCAM system and developed by Richard Pogge at OSU Astronomy Dept. The modifications can be summarized as follows:

- To Replace PC-TCS serial communication with Telcom TCP/IP communication (Comsoft Native protocol → PCTCS-NG Network protocol)
- To add the AUX TCP/IP communication

- To add the auto recovery routine and the communication and links status monitoring
- The telemetry data management for both PCTCS and AUX
- Update and monitoring for the statuses of the telescope and AUX subsystems
- To add high-level remote commands and the processes for the telescope operations (RA/Dec guide, goto, offset, filter change, position init, focus/tip/tilt adjust)
- To extend the runtime configuration for the KMTNet system

In the modification to KMTNet version, first of all, the original RS-232 serial communication link to PC-TCS was replaced with the two TCP/IP communication links to Telcom and AUX server. Main functions were updated accordingly to manage the communication links and the telemetry data for both TCS (PC-TCS and Telcom) and AUX control, to diagnose statuses of the telescope and subsystems, and to process high-level commands for telescope operation.

Both the TCP/IP connection with Telcom and the serial stream of PC-TCS were monitored in the TCS link management. TCS link is set to IDLE in case that the PC-TCS serial stream is invalid. TCS link is set to DOWN in case the TCP/IP connection with Telcom is disconnected or unavailable. In AUX link management, AUX link is set to DOWN if an error is occurred in command send/receive process or the TCP/IP connection with AUX server is disconnected or unavailable. If Auto Recovery mode (ArcMode) is enabled, TCS Agent tries to connect to Telcom/AUX server, and to recover the TCS/AUX links in intervals of ArcInt.

The TCS telemetry data and TCS status monitoring are almost same as the original version. The AUX telemetry data and AUX status monitoring are newly added. In the AUX telemetry data update, TCS Agent monitors the operation statuses of the subsystems and if there was ERROR status or TCS Agent judges to be an abnormal status, TCS Agent prints an error or warning message on the console, and provides the information in the telemetry data string for the remote user.

The high-level commands for telescope operation are processed using combinations of the raw remote commands of PC-TCS and AUX. The PC-TCS remote commands are compliant with "PCTCS-NG Network Protocol" and "COMSOFT Native Protocol" in the document "COMSOFT Legacy PC-TCS Communications", revised by David Harvey of COMSOFT, distributed at the KMTNet software meeting in 2011. The AUX remote commands are compliant with the document "KMTNet Auxiliary control software – Remote commands definition", defined initially by Gilbert Jones, AUX control software developer, and revised by Sang-Mok Cha of KASI.

The runtime configuration was extended for the optimization to KMTNet TCS. New keywords were added to the configuration file to specify Telcom/AUX server information, communication

control settings, and HW configurations. There are more details about in the section about Runtime Configuration.

```

user@localhost:~/TCSagent/KMTNet
File Edit View Search Terminal Help
-----
                KMTNet TCS Agent
      Interactive PC-TCS & AUX Remote Interface Client

      Version: v1.2 (2014-Aug-26 22:48:46)
-----
- Runtime configuration loading completed
- Started TCS Agent as standalone ISIS node TC
  on localhost.localdomain port 6606

- PCTCS Telcom tcp link initialized
- TCS Input Epoch set to 2000.000 success
- AUX ctrl link initialized

- TCS Agent start..

-----
Type 'quit' to terminate TCS Agent process
Type 'help' to see a list of commands
-----
TC% 

```

<KMTNet TCS Agent, executed on the Linux terminal>

3. Installation

3.1. KMTNet TCS Agent package

TCS Agent can be built and run on the system as bellow simple way:

1. Installation of ISIS client library: Login as dts or root user. Copy the source directory of ISIS client library (client) to /home/dts/ISIS/, and execute 'build' script to generate 'libisis.a' file.

```

dts@kmta_ICGui: /home/dts/ISIS/client]# ls
build doc Doxyfile isisclient.h isismessage.c isisserial.c
isissocket.c isisutils.c Makefile
dts@kmta_ICGui: /home/dts/ISIS/client]#

```

```

dts@kmta_ICGui: /home/dts/ISIS/client]# ./build
/bin/rm -f libisis.a *.o
/usr/bin/g++ -c -DISIS_VERSION='"ISISlib v2.0.1"' -DISIS_COMPDATE='"2020-
Nov-10"' -DISIS_COMPTIME='"07:02:00"' isismessage.c
/usr/bin/g++ -c -DISIS_VERSION='"ISISlib v2.0.1"' -DISIS_COMPDATE='"2020-
Nov-10"' -DISIS_COMPTIME='"07:02:00"' isisserial.c
/usr/bin/g++ -c -DISIS_VERSION='"ISISlib v2.0.1"' -DISIS_COMPDATE='"2020-
Nov-10"' -DISIS_COMPTIME='"07:02:00"' isissocket.c
/usr/bin/g++ -c -DISIS_VERSION='"ISISlib v2.0.1"' -DISIS_COMPDATE='"2020-
Nov-10"' -DISIS_COMPTIME='"07:02:00"' isisutils.c
/bin/rm -f libisis.a
/usr/bin/ar cr libisis.a isismessage.o isisserial.o isissocket.o
isisutils.o
ranlib libisis.a
dts@kmta_ICGui: /home/dts/ISIS/client]#
dts@kmta_ICGui: /home/dts/ISIS/client]# ls
build doc Doxyfile isisclient.h isismessage.c isisserial.c
isissocket.c isisutils.c libisis.a Makefile

```

2. Execute 'build' script in the source directory of TCS Agent,
/home/kasi/TCSAgent/KMTNet/. Then the executable file 'pctcs' will be made.
3. Type "\$./pctcs ./pctcs.ini" to run with example/default runtime config file in the same directory.

TCS Agent package include 10 files as bellow:

- 00README.txt : directions for 1st building and running the TCS Agent
- pctcs.h : main TCS Agent header file
- main.c : main func for interface monitoring and processing
- loadconfig.c : load an TCS Agent's runtime configuration file
- comsoft.c : ComSoft PCTCS and AUX control utility routines
- commands.h : command tree header for the TCS Agent
- commands.c : command handler, command action, and utility functions
- pctcs.ini : runtime configuration file (default)
- Makefile : compile and build configuration
- build (exec) : script to build a executable file of TCS Agent
- pctcs (exec) : executable file of TCS Agent, built by the script 'build'

The TCS Agent's executable file 'pctcs' can be built using the shell script 'build' for compilation information. The version information is revised in the VERSION definition of 'Makefile'. If the VERSION is not defined in the 'Makefile' the version information that is defined in 'pctcs.h' will be used in the compile.

Both the ISIS client library of OSU and GNU Readline library are required to build a TCS Agent. The ISIS client library must be installed in the directory `"/home/dts/ISIS/client"` before building TCS Agent. `"isisclient.h"` and `"libisis.a"` files are used in the compile. They can be built using the script 'build' in the same directory. The GNU Readline library must be installed in the system on that TCS Agent will be running, before building TCS Agent. The Readline installation files with open source and detailed manuals can be downloaded in the Readline homepage.

If TCS Agent is executed without the argument for runtime configuration file, the default name and location setting `"/home/dts/Config/pctcs.ini"` will be used. If the user wants to use the default, a `pctcs.ini` should be copied to the default location. If the user does not want to use the default name and location, the user can put a arguments for a user-defined file name with the path, e.g. `"$ ./pctcs ./pctcs.ctio.ini"`.

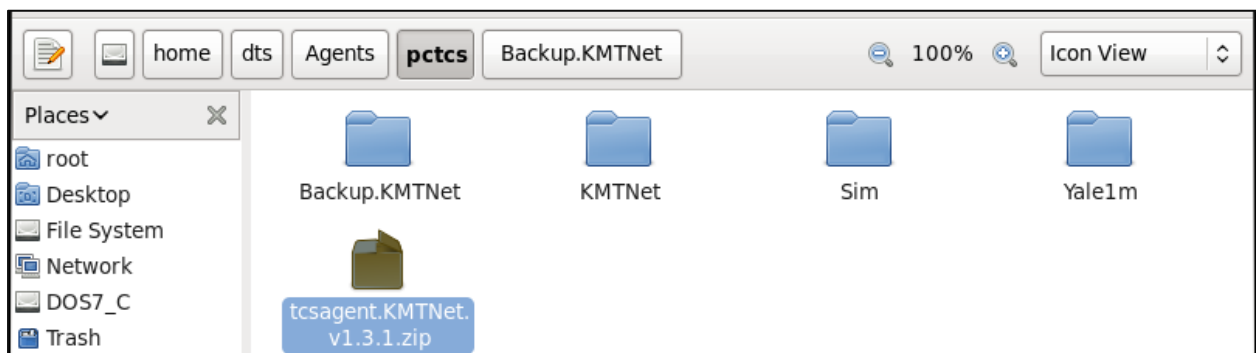
3.2. KMTNet TCS Agent installation on the ICS

The KMTNet TCS Agent is installed on the ICS (Instrument Control & Science Data Acquisition Control) server. The system maintainer of KASI installs the new version of KMTNet TCS Agent into the `"/home/dts/Agents/pctcs/KMTNet"` and `"/home/dts/bin"` when it is updated. The new runtime configuration file is also replaced with the new version files in the `"/home/dts/Config"` directory, default location of the runtime configuration file, if it was updated. ICIMACS use the binary file in the `"/home/dts/bin/pctcs"` to launch all clients of ICIMACS. Following guide provides how to install the KMTNet TCS Agent on the ICS.

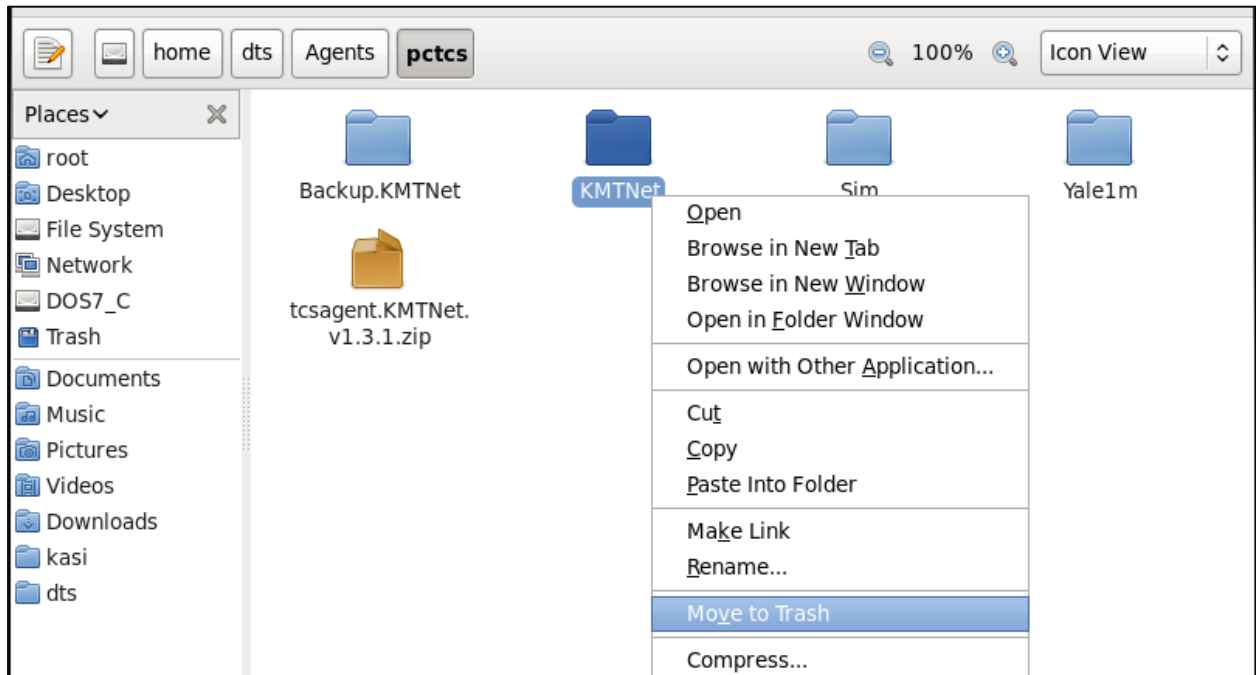
KMTNet TCS Agent Installation Guide>>

A. INSTALLATION GUIDE :

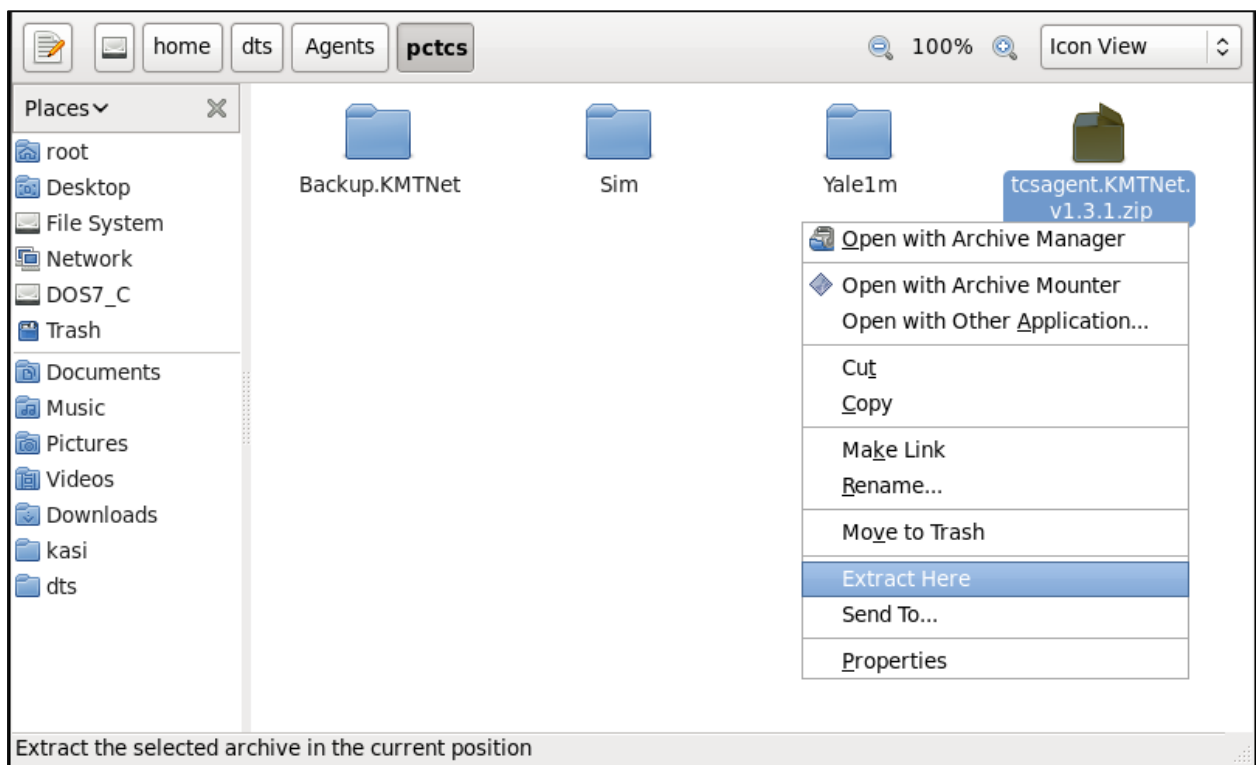
1) Copy a compressed file of KMTNet TCS Agent new version into `"/home/dts/Agents/pctcs"` directory. The file name is generally `'tcsagent.KMTNet.v#.#.#'`.

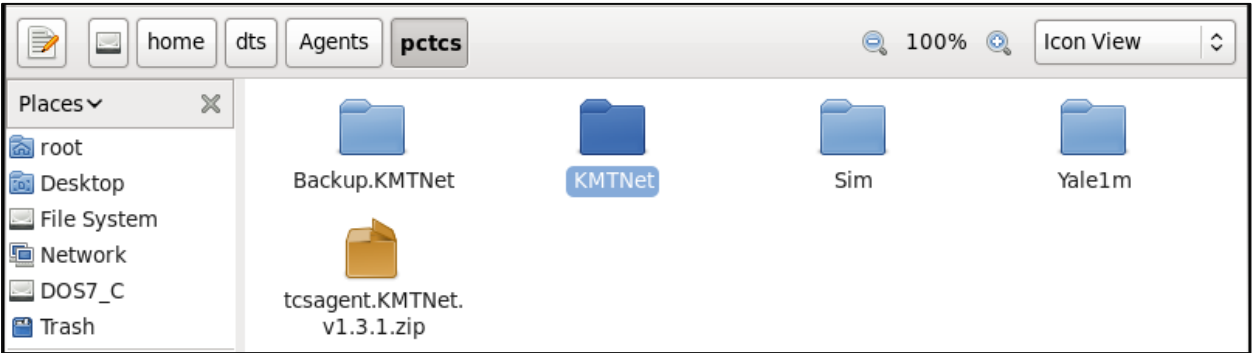


- 2) Remove the old KMTNet TCS Agent directory, named 'KMTNet', in the '/home/dts/Agents/pctcs' directory.

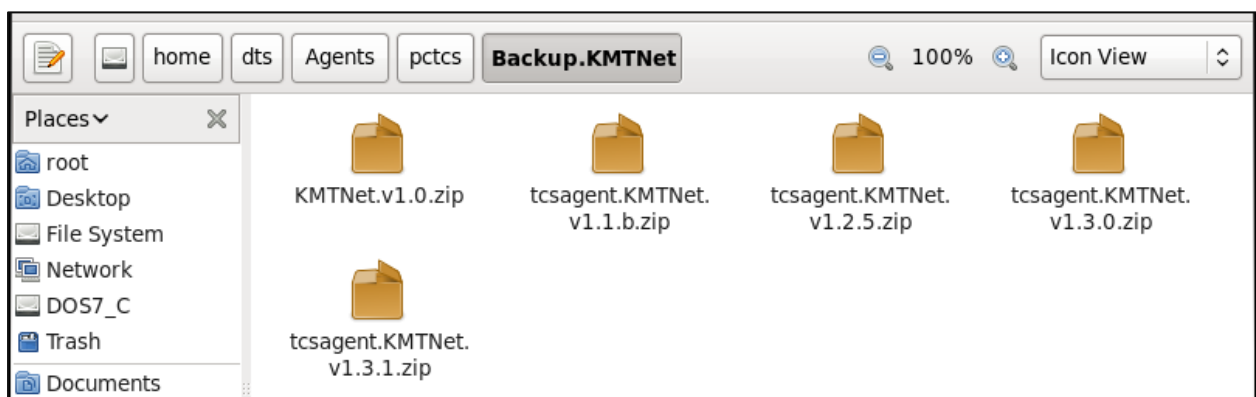
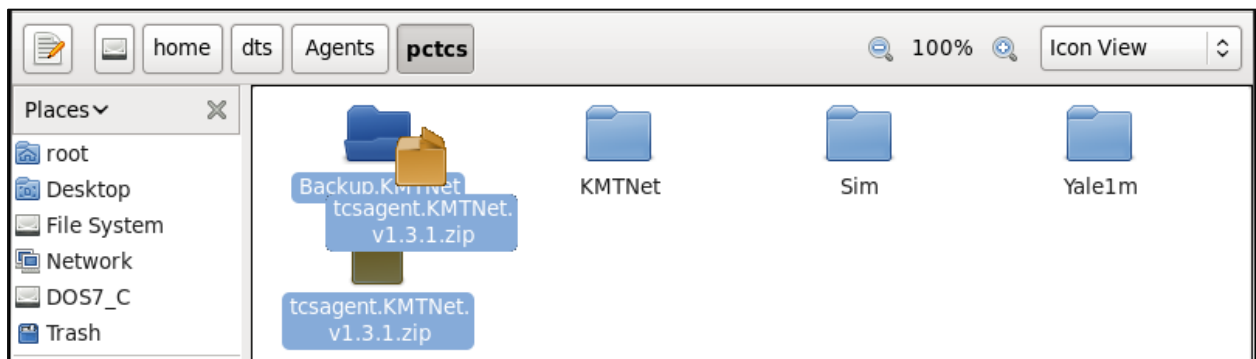


- 3) Extract the new KMTNet TCS Agent directory from the compressed file in the '/home/dts/Agents/pctcs' directory. Then new 'KMTNet' directory will be generated.





- 4) Move the compressed file into the 'Backup.KMTNet' directory. The compressed files of old versions are stored for backup in the directory.



- 5) Build a binary file 'pctcs' of the new TCS Agent using the 'build' script in the KMTNet TCS Agent directory.

```

root@KMTN17:/home/dts/Agents/pctcs/KMTNet
File Edit View Search Terminal Help
[root@KMTN17 ~]# cd /home/dts/Agents/pctcs/
[root@KMTN17 pctcs]# ls -al
total 24
drwxrwxr-x. 6 dts dts 4096 Sep  5 22:23 .
drwxr-xr-x. 8 dts dts 4096 Apr 12 2013 ..
drwxr-xr-x  2 root root 4096 Sep  5 22:23 Backup.KMTNet
drwxrwxr-x  2 root root 4096 Sep  5 19:59 KMTNet
drwxrwxr-x. 2 dts dts 4096 Feb 27 2014 Sim
drwxrwxr-x. 2 dts dts 4096 Nov  8 2013 Yale1m
[root@KMTN17 pctcs]# cd KMTNet/
[root@KMTN17 KMTNet]# ls -al
total 244
drwxrwxr-x  2 root root  4096 Sep  5 19:59 .
drwxrwxr-x. 6 dts dts  4096 Sep  5 22:23 ..
-rw-rw-r--  1 root root 2864 Aug 26 22:53 00README.txt
-rwxrwxrwx  1 root root  141 May 27 2005 build
-rw-rw-rw-  1 root root 104462 Sep  5 19:59 commands.c
-rw-rw-rw-  1 root root  5927 Sep  2 14:25 commands.h
-rw-rw-rw-  1 root root 21255 Sep  2 17:13 comsoft.c
-rw-rw-rw-  1 root root 27665 May  3 22:36 loadconfig.c
-rw-rw-rw-  1 root root 29345 Sep  2 16:11 main.c
-rw-rw-rw-  1 root root  1207 Aug  4 20:45 Makefile
-rw-rw-rw-  1 root root 22175 Sep  5 19:56 pctcs.h
-rw-rw-rw-  1 root root  2177 Aug 31 15:32 pctcs.ini
[root@KMTN17 KMTNet]# ./build
/bin/rm -f pctcs *.o
/usr/bin/g++ -w -c -I/home/dts/ISIS/client -L/home/dts/ISIS/client -DAPP_VERSION="" -DAPP_COMPDATE="2014-Sep-05" -DAPP_COMPTIME="22:29:19" loadconfig.c
/usr/bin/g++ -w -c -I/home/dts/ISIS/client -L/home/dts/ISIS/client -DAPP_VERSION="" -DAPP_COMPDATE="2014-Sep-05" -DAPP_COMPTIME="22:29:19" commands.c
/usr/bin/g++ -w -c -I/home/dts/ISIS/client -L/home/dts/ISIS/client -DAPP_VERSION="" -DAPP_COMPDATE="2014-Sep-05" -DAPP_COMPTIME="22:29:19" comsoft.c
/usr/bin/g++ -o pctcs -DAPP_VERSION="" -DAPP_COMPDATE="2014-Sep-05" -DAPP_COMPTIME="22:29:19" main.c loadconfig.o commands.o comsoft.o -I/home/dts/ISIS/client -L/home/dts/ISIS/client -lisis -lreadline -lhistory -lncurses -lm
[root@KMTN17 KMTNet]#
[root@KMTN17 KMTNet]# ls -al

```

6) Copy the binary file 'pctcs' to the '/home/dts/bin' directory, and

Copy the runtime configuration file 'pctcs.ini' to the '/home/dts/Config' directory or replace old 'pctcs.ini' and 'pctcs.kmtcn' files with the new file in the directory.

```

root@KMTN17:/home/dts/Agents/pctcs/KMTNet
File Edit View Search Terminal Help
[root@KMTN17 KMTNet]# ls -al
total 340
drwxrwxr-x  2 root root   4096 Sep  5 22:29 .
drwxrwxr-x  6 dts dts    4096 Sep  5 22:23 ..
-rw-rw-r--  1 root root   2864 Aug 26 22:53 00README.txt
-rwxrwxrwx  1 root root    141 May 27  2005 build
-rw-rw-rw-  1 root root 104462 Sep  5 19:59 commands.c
-rw-rw-rw-  1 root root   5927 Sep  2 14:25 commands.h
-rw-rw-rw-  1 root root  21255 Sep  2 17:13 comsoft.c
-rw-rw-rw-  1 root root  27665 May  3 22:36 loadconfig.c
-rw-rw-rw-  1 root root  29345 Sep  2 16:11 main.c
-rw-rw-rw-  1 root root   1207 Aug  4 20:45 Makefile
-rwxr-xr-x  1 root root  97394 Sep  5 22:29 pctcs
-rw-rw-rw-  1 root root  22175 Sep  5 19:56 pctcs.h
-rw-rw-rw-  1 root root   2177 Aug 31 15:32 pctcs.ini
[root@KMTN17 KMTNet]#
[root@KMTN17 KMTNet]# cp pctcs /home/dts/bin/
cp: overwrite `/home/dts/bin/pctcs'? y
[root@KMTN17 KMTNet]# cp pctcs.ini /home/dts/Config/pctcs.ini
cp: overwrite `/home/dts/Config/pctcs.ini'? y
[root@KMTN17 KMTNet]# cp pctcs.ini /home/dts/Config/pctcs.kmtcn
cp: overwrite `/home/dts/Config/pctcs.kmtcn'? y
[root@KMTN17 KMTNet]# chmod 666 /home/dts/Config/pctcs.ini
[root@KMTN17 KMTNet]# chmod 666 /home/dts/Config/pctcs.kmtcn
[root@KMTN17 KMTNet]# ls -al /home/dts/Config/pctcs.*
-rw-r--r--  1 dts dts   886 Jul  1 09:31 /home/dts/Config/pctcs.direct
-rw-rw-rw-  1 dts dts  2177 Sep  5 22:50 /home/dts/Config/pctcs.ini
-rw-rw-rw-  1 dts dts  2177 Sep  5 22:51 /home/dts/Config/pctcs.kmtcn
[root@KMTN17 KMTNet]#
[root@KMTN17 KMTNet]#

```

7) Run the KMTNet TCS Agent with '/home/dts/bin/pctcs' for check.

```

root@KMTN17:~
File Edit View Search Terminal Help
[root@KMTN17 ~]# /home/dts/bin/pctcs

-----
                KMTNet TCS Agent
    Interactive PC-TCS & AUX Remote Interface Client

    Version: v1.3.1 (2014-Sep-05 22:29:19)
-----

- Runtime configuration loading completed
- Started TCS Agent as standalone ISIS node TC
  on KMTN17.ctio.noao.edu port 6606

- PCTCS Telcom tcp link initialized
- TCS Input Epoch set to 2000.000 success
- AUX ctrl link initialized

- TCS Agent start..

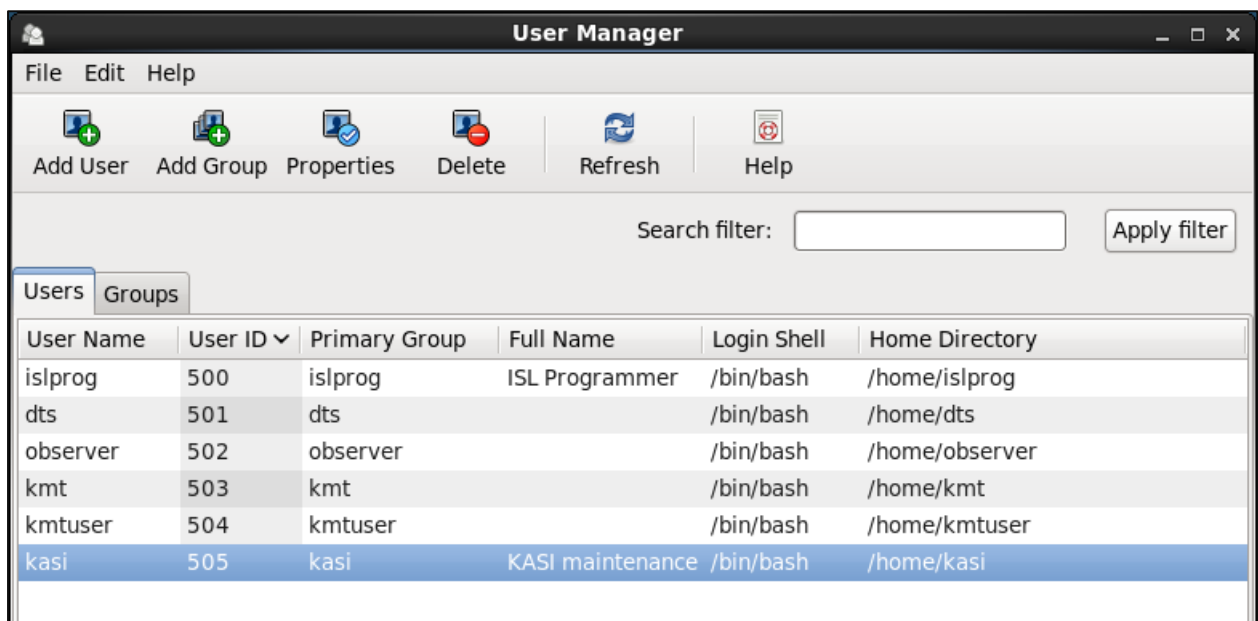
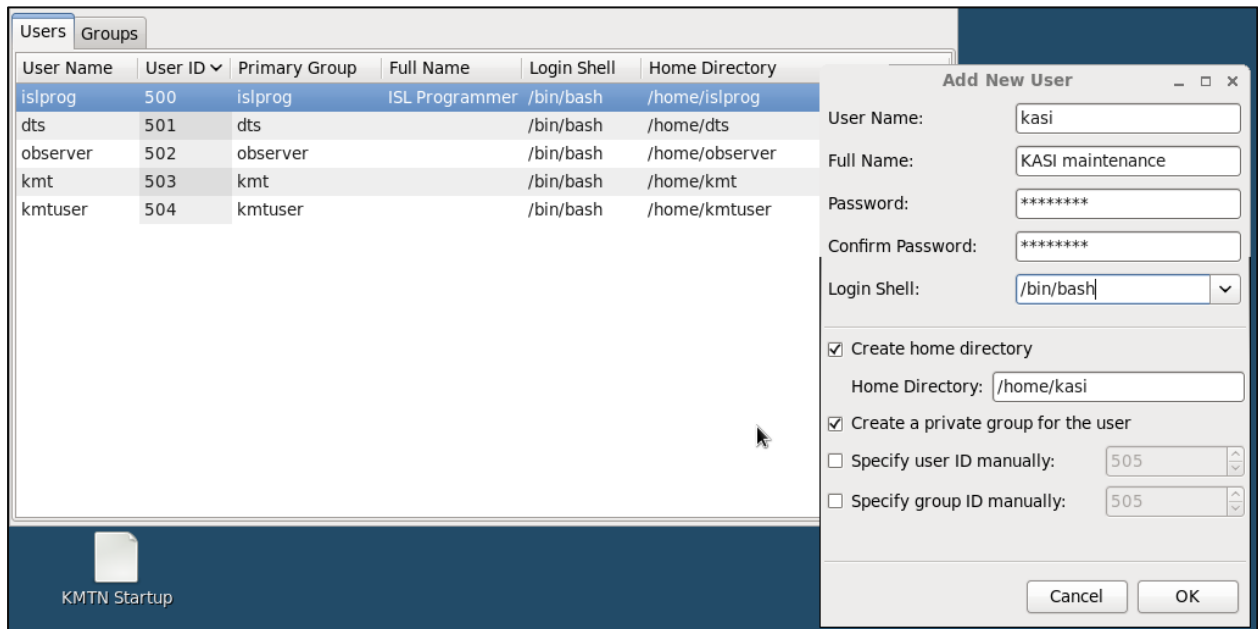
-----
Type 'quit' to terminate TCS Agent process
Type 'help' to see a list of commands
-----

TC% version
DONE: KMTNet TCS Agent Version=(v1.3.1) CompileDate=2014-Sep-05 CompileTime=22:2
9:19
TC% info
DONE: KMTNET TCS Agent v1.3.1 ID=TC Host=KMTN17.ctio.noao.edu:6606 Mode=STANDALO
NE TCSSHost=192.168.14.60:5750 TCSTelID=KMTNET TCSSysID=TCS TCSLink=Up PctcsTime
out=5 sec TelcomTimeout=8 sec TcsUpdateInt=1.0 sec TcsGuideStepRA=0.00503608 arc
sec/encount TcsGuideStepDec=0.01321939 arcsec/encount TcsGuideMinOffsetRA=0.01 a
rcsec TcsGuideMinOffsetDec=0.02 arcsec TcsArcMode=Enabled AUXHost=192.168.14.60:
5752 AUXTelID=KMTNET AUXSysID=AUX AUXLink=Up AuxUpdateInt=0.2 sec AuxFilterOpTim
e=15.0 sec AuxShutOpTime=5.0 sec AuxFAnSouth=2 AuxFAnEast=1 AuxFAnWest=3 AuxArcM
ode=Enabled ArcInt=1.5 Concise -DEBUG -DOLLOG rcfile=/home/dts/Config/pctcs.ini e
xe=/home/dts/bin/pctcs UserID=root Start=2014-09-06T02:54:10
TC% █

```

B. NOTES:

1) Additional account was created on the ICS for KMTNet TCS Agent installation and maintenance. The ID/PW is 'kasi' and 'kasimain'.



2) When the KMTNet TCS Agent is updated, it will be re-installed on the ICS by the system maintainer of KASI, according to this guide. Therefore the KMTNet TCS Agent and its runtime configuration files in 'dts/Agents/pctcs/KMTNet', 'dts/Config' and 'dts/bin' directory of the ICS will be maintained to latest version.

3) The KMTNet TCS Agent is launched by KMTN_Startup script with the binary file 'dts/bin/pctcs'.

4. Runtime Configuration

Runtime configuration is accomplished using an external runtime configuration file (e.g., pctcs.ini) to specify Telcom/AUX server information, communication control settings, application settings, and HW configurations. Basic commands allow some dynamic re-configuration for application setting, but generally the configuration file is given primacy for critical parameters such as the server information and the HW configuration.

The runtime configuration file of KMTNet TCS Agent was extended based on the original runtime configuration file of ISIS client. The basic format is same as typical ISIS-style runtime configuration files. The files contain simple Keyword-Value pairs that are parsed into global-scope variables for the client and its various subroutines to use. The Keyword-Value pair can be intentionally omitted. If a keyword is not defined in the file, the value for the keyword is replaced with the default value that is defined in the source code, maybe 'pctcs.h'.

The '#' is used as a comment character, making a comment line when it appears as the first character in a line by itself. Inline comments are not supported by this simple parser, but are generally ignored since it assumes (again for simplicity) that value arguments are numbers or strings without spaces. Fancier parsers can be implemented as needed, such as the comment is to

the end of the line. Blank lines are ignored by the parser. We also follow the convention that keywords and values are case insensitive, to remove any ambiguity.

The following example is a template of the runtime configuration file of KMTNet TCS Agent. All values of the example were filled in with the defaults, defined in 'pctcs.h', for reference.

< Runtime Configuration File of KMTNet TCS Agent (defaults) >

```
#
# KMTNet TCS Agent runtime config file
#
# Note that while mixed cases are used for readability, entries
# are case insensitive *EXCEPT* for network hostnames or Unix
# filenames. Beware!
#
# R. Pogge, OSU Astronomy Dept.
# pogge@astronomy.ohio-state.edu
# 2005 May 27 (Original version - Yale1m)
#
# S. Cha, KASI KMTNet team
# chasm@kasi.re.kr
# 2014 April 1 (KMTNet version)
#
# - System: site name
# - last update: 2014-MM-DD
#
#####

# Application Mode: either STANDALONE or ISISclient
Mode Standalone
#Mode ISISclient

# ISIS Server Info - only releveant if Mode=ISISclient
ISISID    IS
ISISHost  localhost
ISISPort  6600

# TCS Agents client info (Host=localhost is implicit)
ID    TC
Port  6606

# TCS server (PCTCS Telcom) info
TCS_Host xxx.xxx.xxx.xxx (put here the IP and a comment for the site name)
TCS_Port 5750
TCS_TelID KMTNET
TCS_SysID TCS

# AUX control server info
AUX_Host xxx.xxx.xxx.xxx (put here the IP and a comment for the site name)
AUX_Port 5752
AUX_TelID KMTNET
AUX_SysID AUX

# Timeout setting for Telcom tcp and PC-TCS serial link (in second)
Timeout_PCTCS 5
Timeout_Telcom 7
```

```

# TCS & AUX telemetry update interval (in second)
UpdateInt_TCS  1.0
UpdateInt_AUX  0.2

# TCS & AUX TCP Links auto recovery mode setting (on/off)
AutoRecovery_TCS  on
AutoRecovery_AUX  on

# TCS & AUX TCP Links auto recovery try interval (in seconds)
ArcInt  2.0

# HW configuration
TCS_Guide_Step_RA      0.00503608  (in arcsec/encoder_count)
TCS_Guide_Step_Dec     0.01321939  (in arcsec/encoder_count)
TCS_Guide_MinOff_RA    0.01        (in arcsec, minimum RA guiding offset)
TCS_Guide_MinOff_Dec   0.02        (in arcsec, minimum Dec guiding offset)
AUX_FS_Filter_OpTime   20.0        (in seconds, filter slide operation time)
AUX_FS_Shutter_OpTime  12.0        (in seconds, camera shutter opening time)
AUX_FA_ActNum_South    2           (south actuator number)
AUX_FA_ActNum_East     1           (east actuator number)
AUX_FA_ActNum_West     3           (west actuator number)

# TCS Agent client application runtime flags (on/off)
VERBOSE  off
DEBUG    off
DOLOG    off

# TCS Agent log file path
LOGFILE  "/data/Logs/pctcs.log"  (default)

# END

```

The MODE keyword is for application mode. TCS Agent is configured either as an ISIS client, or as a standalone program operating independently of an ISIS system. In the Standalone mode, TCS Agent accepts and processes commands from any client that complies with IMPv2. In the ISIS client mode, TCS Agent interacts with only ISIS that has the ISISID and the ISISPORT, and also an ISIS client terminal is enabled so the local user can interact with ISIS on the console.

In the ISIS server information, ISISID is Node name of the ISIS server, used in ICIMACS. ISISHOST is the network hostname of the machine running the ISIS server. It can be set with a network hostname or an IP address. ISISPORT is the network (UDP) socket's port number used by the ISIS server running on ISISHOST. These are only meaningful if MODE ISISCLIENT has been set. In the TCS Agents client information, ID is the node name of TCS Agent, used in ICIMACS. PORT is the network (UDP) socket's port number used by TCS Agent to listen to ICIMACS server's commands.

In the TCS server (PCTCS Telcom) and AUX server information, TCS/AUX_HOST is the network

hostname or IP address of the machine running the Telcom/AUX server. Using IP address is more robust to find the host because there ICIMACS including TCS Agent and KMTNet TCS including Telcom/AUX server are in different subnets normally. PORT is TCP socket's port number of the Telcom/AUX server. TELID and SYSID are used for raw remote command of PC-TCS Telcom and AUX control. These are set and fixed in the telescope system installation so they will not be changed in each site.

TIMEOUT_PCTCS is a timeout setting used for monitoring the PC-TCS serial link. If PC-TCS telemetry data is invalid and PC-TCS is judged to be idle for the timeout, the TCS link status become IDLE. TIMEOUT_TELCOM is a timeout setting used for monitoring the Telcom TCP link. If the TCP communication with Telcom is disconnected or unavailable for the timeout, the TCP link status become DOWN. There is no a timeout setting for AUX link because monitoring the AUX link can be performed every telemetry update in a short interval.

UPDATEINT_TCS/AUX is an interval for the TCS/AUX telemetry and status update. TCS Agent allows a minimum of 0.5 seconds for TCS update interval and a minimum of 0.1 seconds for AUX update, according to the specifications for Telcom/AUX server. AUTORECOVERY_TCS/AUX is a flag to set the auto recovery mode (ArcMode) for TCS/AUX link. ArcMode can be re-configured using commands on the prompt. ARCINT is an interval of the auto recovery try for both TCS link and AUX link. It is allowed a minimum 0.5 seconds. If ArcMode is enabled, TCS Agent tries to connect to Telcom/AUX server, and to recover the TCS/AUX links in intervals of ArcInt when the link status is DOWN.

In the HW configuration section, TCS_GUIDE_STEP_RA/DEC is a HW specification of KMTNet driving system. It is arc-seconds per encoder count on RA/Dec, and used for guiding offset operation. The values may be identical at all KMTNet site if HW specification is not changed. TCS_GUIDE_MINOFF_RA/DEC is a setting for guiding-offset operation. In the command processing, if commanded guiding offset is less than this value, TCS Agent does not process the guide offset command for each axis to reduce overhead. AUX_FS_FILTER/SHUTTER_OPTIME is a HW specification for the operation time of the filter or shutter slide. This value is used to diagnose the operation status of the filter or shutter. It can be changed according to the HW setting of Filter/Shutter assembly. AUX_FA_ACTNUM_SOUTH/EAST/WEST is the actuator number for each orientation and used for the tip/tilt adjustment. This setting depends on the actuator installation condition and AUX control software setting.

VERBOSE, DEBUG and DOLOG are flags for the application setting, and re-configured using commands on the prompt. The function for runtime logging is not implemented in the current version.

5. Commands

5.1. Command list

KMTNet TCS Agent provides the command sets for client setting, TCS (PC-TCS and Telcom) control and AUX control. The client command set includes commands for application information, setting and control. The TCS command set includes commands for TCS link control, the telescope mount operation, and the TCS status/telemetry data. The AUX command set includes commands for AUX link control, the subsystems control, and the AUX status/diagnostic data/telemetry data. All commands are used as keyboard command by the local user on the prompt console, and also as remote command by the remote host via the IMP-compliant interface. The EXEC-type commands are generally not allowed for remote operation, but it can be executed by remote command with an explicit EXEC: command request. But the remoter operation of EXEC-type command is not recommended if not the purpose of system maintenance. The EXEC-type commands can be executed by remote command with an explicit EXEC: command request. But generally it is not recommended if not the purpose of system maintenance. The EXEC-type commands is always allowed for keyboard command. The following list shows the commands of KMTNet TCS Agent with information for arguments, format and message type.

<KMTNet TCS Agent interactive commands list>

Set	Command and Arguments	Arguments format [unit] / <return>	Msg.Type
Client	quit	-	EXEC
	init	-	REQ
	reset	-	
	close	-	
	arc	-	
	info	-	
	version	-	
	verbose	-	
	debug	-	
	history	<commands history>	EXEC
	!!	-	
	!cmd	-	
	help or ?	<commands list and descriptions>	

TCS	tcsinit	-	REQ
	tcsreset	-	
	tcsclose	-	
	tcsarc	-	
	tcsstatus	<TCS status in IMPv2>	
	tstat	<TCS status in lightweight>	
	traw	<raw telemetry data string>	
	tsync	-	EXEC
	tcmd <tcmd>	(COMSOFT Native Protocol)	REQ
	tguide <ra_offset> <dec_offset>	$\pm x.xx \pm x.xx$ [arcsec]	
	tgoto <ra> <dec>	hh:mm:ss.sss $\pm$ dd:mm:ss.ss [J2000]	
	toffset <ra_offset> <dec_offset>	$\pm$ hh:mm:ss.ss $\pm$ dd:mm:ss.s	
	tstop	-	
	tdi	-	
AUX	auxinit	-	REQ
	auxreset	-	
	auxclose	-	
	auxarc	-	
	auxstatus	<AUX status in IMPv2>	
	astat	<AUX status in lightweight>	
	acmd <acmd>	(AUX control remote commands)	
	filter <fnum>	0 ~ 4 : filter number	
	fsastat	<Filter/Shutter status in lightweight>	
	dfocus <dfoc>	$\pm x.xxx$ [mm] : Δ focus	
	dtilt <dtns> <dtew>	$\pm x.x \pm x.x$ [aresec] : Δ tilt angle	EXEC
	fttgoto <foc> (<tns> <tew>)	$\pm x.xxx (\pm x.x \pm x.x)$ ["] : abs. positions	REQ/(EXEC)
	fttstat	<Focus/Tip-tilt status in lightweight>	REQ

5.2. Command details

The telescope operating commands are processed using combinations of the raw commands of PC-TCS and AUX. The raw PC-TCS commands and the raw AUX commands are according to "PCTCS-NG Network Protocol" and "KMTNet Auxiliary control software – Remote commands definition". If the verbose output mode is enabled, we can see the raw commands input/output, processed by low-level routines.

The command processes do not monitor the status in the operation, and do not check the final result for the operation. The operation commands return the completion acknowledgement with 'DONE' message, if the raw command was just sent without error. So the final result for the

operation should be checked with the status and telemetry data by kinds of STATUS command.

The following sections describe the commands with usage and notes. The argument format is comprised in Usage with the recommended significant figures using 'x' symbols. It reflects the specification of KMTNet TCS. Details of the status and telemetry data are described in section (4).

(1) Client commands

QUIT quit TCS Agent application

(Usage) quit

(Details) allowed only if EXEC from remote hosts

(Notes)

INIT initialize both TCS & AUX links

(Usage) init

(Details)

(Notes)

RESET reset/restart both TCS & AUX links

(Usage) reset

(Details) alias for INIT

(Notes)

CLOSE close both TCS & AUX links

(Usage) close

(Details)

(Notes)

ARC toggle the auto recovery mode for both TCS & AUX links

(Usage) arc

(Details)

(Notes)

INFO report the application information

(Usage) info

(Details) This command returns the application information and settings.

(Notes)

VERSION report the application version and compilation information

(Usage) version

(Details)

(Notes)

VERBOSE toggle the verbose output mode

(Usage) verbose

(Details)

(Notes)

DEBUG toggle the debugging message output mode

(Usage) debug

(Details)

(Notes)

HISTORY show the command history list

(Usage) history

(Details) allowed only if EXEC from remote hosts

(Notes)

!! - repeat the last command

(Usage) !!

(Details) allowed only if keyboard command

(Notes)

!CMD repeat the last command matching 'CMD'

(Usage) !cmd

(Details) allowed only if keyboard command

(Notes)

HELP / ? view the TCS Agent commands list and simple descriptions

(Usage) help / ?

(Details) allowed only if EXEC from remote hosts

(Notes)

(2) TCS commands

TCSINIT initialize TCS (PC-TCS Telcom) link

(Usage) tcsinit

(Details)

(Notes)

TCSRESET reset/restart TCS (PC-TCS Telcom) link

(Usage) tcsreset

(Details) alias for TCSINIT

(Notes)

TCSCLOSE close TCS (PC-TCS Telcom) link

(Usage) tcsclose

(Details)

(Notes)

TCSARC toggle the auto recovery mode for TCS link

(Usage) tcsarc

(Details) If Enabled, TCS Agent will try to connect to Telcom server and to recover TCS link at an interval of ArcInt (auto recovery interval) defined the runtime configuration file (pctcs.ini)

(Notes)

TCSSTATUS query & return the TCS status and telemetry data in IMPv2

(Usage) tcsstatus

(Details) The last-updated telemetry data are returned, or just the time/date info and ARC mode setting are returned if the TCS link is down or idle too long. This command returns TCS status and telemetry data as a valid IMPv2 message string. Refer to section (4) for more details of the TCS status and telemetry data

(Notes) This is usually within 20 msec of the query, so the lag is small. But the update interval for TCS telemetry is typically 1 sec, so the data can be stuff acquired 1 sec ago(maximum).

TSTAT query & return the raw TCS status and telemetry data without keywords

(Usage) tstat

(Details) lightweight TCSSTATUS

The last-updated telemetry data are returned, or just the time/date info and ARC mode setting are returned if the TCS link is down or idle too long. This command returns a lightweight status string in simple, non-IMPv2-compliant format for simple reading/parsing by machines not humans. The format is as follows, depending on the TCS link state:

TCS_UP: PC-TCS link active

UP DATE-OBS TIME-OBS TIMESYS DATE-UP TIME-UP ARCMODE
RA DEC EQUINOX HA ST SECZ ALT AZ MOVESTATUS EXECODE

TCS_IDLE: PC-TCS link has been idle for longer than the allowed time

IDLE DATE-OBS TIME-OBS TIMESYS ARCMODE

TCS_DOWN: PC-TCS link is disabled ("down")

DOWN DATE-OBS TIME-OBS TIMESYS ARCMODE

Refer to section (4) for more details of the raw TCS status and telemetry data

(Notes) This is usually within 20 msec of the query, so the lag is small. But the update interval of TCS telemetry is typically 1 sec, so the data can be stuff acquired 1 sec ago(maximum).

TRAW return the raw PC-TCS telemetry string of the latest packet

(Usage) traw

(Details) The raw PC-TCS telemetry string is returned. This command may be used for PC-TCS maintenance or debugging TCS Agent.

(Notes)

TSYNC synch the PC-TCS clock with the local system clock

(Usage) tsync

(Details) allowed only if EXEC from remote hosts

(Notes) User must check the UT date will not change soon before using this command. If the time is pass on 24:00 in progress, the date will be not correct since time-sync command and date-sync command are separated and there is some delay in the process.

TCMD send a raw PC-TCS remote command

(Usage) tcmd <tcmd>

<tcmd>: PC-TCS raw commands, consist of <native cmd> <argument>

(Details) PC-TCS raw commands are defined in "COMSOFT Native Protocol" of the "COMSOFT Legacy PC-TCS Communications" document. <tcmd> doesn't include TELID, SYSID and PID defined in "PCTCS-NG Network Protocol"

(Notes) This command returns the completion acknowledgement with 'DONE' message, if the raw command was just sent without error. The final result for the operation should be checked with the status and telemetry data by TSTAT or TCSSTATUS.

Refer to section (4) for more details of the raw AUX status and telemetry data

TGUIDE move the telescope as RA/Dec guiding offset

(Usage) tguide <RA_offset> <Dec_offset>

<RA_offset>: RA guiding offset, +x.xx (arcsec), angular distance (NOT delta RA)

<Dec_offset>: Dec guiding offset, +x.xx (arcsec)

(Details) The process is separated for RA guiding offset and Dec guiding offset. The RA guiding offset is commanded, and then Dec guiding offset is commanded.

(Notes) If the RA/Dec offset value is less than TCS_Guide_MinOff_RA/Dec, TCS Agent simply ignores the command for each axis. For example, If the RA offset value is

more than TCS_Guide_MinOff_RA and the Dec offset value is less than TCS_Guide_MinOff_Dec, only the RA axis will move as guiding offset.

TGOTO goto J2000 RA/Dec

(Usage) tgoto <RA> <Dec>

<RA>: hh:mm:ss.sss (J2000)

<Dec>: ±dd:mm:ss.ss (J2000)

(Details)

(Notes) TCS Agent sets the Input Epoch of PC-TCS to J2000 automatically at TCS link initialization. But when PC-TCS is not running or not in the normal operation, the automatic Input Epoch setting can be not successful and TCS Agent displays a message for the Input Epoch setting failure. If the Input Epoch setting is not initialized successfully, the Input Epoch must be set to J2000 manually on PC-TCS before commanding 'TGOTO' or 'TOFFSET' to slew the telescope.

The final result of this operation should be checked with the status and telemetry data by TSTAT or TCSSTATUS.

TOFFSET offset move the telescope for RA/Dec

(Usage) toffset <RA_offset> <Dec_offset>

<RA_offset>: RA component of the offset vector, ±hh:mm:ss.ss, delta RA

<Dec_offset>: Dec component of the offset vector, ±dd:mm:ss.s, delta Dec

(Details)

(Notes) The final result of this operation should be checked with the status and telemetry data by TSTAT or TCSSTATUS.

TSTOP cancel command and stop the telescope's motion

(Usage) tstop

(Details)

(Notes) This command is useful in software interface test performed without the telescope.

TDI synch the current position with the commanded position

(Usage) tdi

(Details) This command synchronizes the telescope by forcing the current position to become the same as the commanded position.

(Notes) This command is useful for software interface test performed without the telescope.

(3) AUX commands**AUXINIT** initialize AUX link

(Usage) auxinit

(Details)

(Notes)

AUXRESET reset/restart AUX link

(Usage) auxreset

(Details) alias for AUXINIT

(Notes)

AUXCLOSE close AUX link

(Usage) auxclose

(Details)

(Notes)

AUXARC toggle the auto recovery mode for AUX link

(Usage) auxarc

(Details) If Enabled, TCS Agent will try to connect to AUX control remote server and to recover AUX link at an interval of ArcInt (auto recovery interval) defined the runtime configuration file (pctcs.ini).

(Notes)

AUXSTATUS query & return the AUX status and telemetry data in IMPv2

(Usage) auxstatus

(Details) The last-updated telemetry data are returned, or just the time/date info and ARC mode setting are returned if the AUX link is down. This command returns AUX status and telemetry data as a valid IMPv2 message string.

Refer to section (4) for more details of the AUX status and telemetry data

(Notes) This is usually within 20 msec of the query, so the lag is small. But the update interval for AUX telemetry is typically 200 msec, so the data can be stuff acquired 0.2 sec ago(maximum).

ASTAT query & return raw AUX status without keywords

(Usage) astat

(Details) lightweight AUXSTATUS,

The last-updated status, diagnostic data and telemetry data are returned, or just the time/date info and ARC mode setting are returned if the AUX link is down. This command returns a lightweight status string in simple, non-IMPv2-

compliant format for simple reading/parsing by machines not humans.

The format is as follows, depending on the AUX link state:

AUX_UP:

```
UP DATE-OBS TIME-OBS TIMESYS DATE-UP TIME-UP ARCMODE
FS: FSSTAT FOPSTAT FNUMBER SOPSTAT SSTATUS
FA: FASTAT FOCUS TILTNS TILTEW SOUTHLIM EASTLIM WESTLIM
    SOUTHPOS EASTPOS WESTPOS
DS: DSSTAT UPSHUT LOSHUT SAFSW AUTOSYNC SHUTALT TELEALT
MC: MCSTAT MCPOS
CH: CHSTAT COOLING SETPOINT PROCTEMP
EN: ENSTAT FAN S1 S2 S3 S4 S5 S6 S7
```

AUX_DOWN:

```
DOWN DATE-OBS TIME-OBS TIMESYS ARCMODE
```

Refer to section (4) for more details of the raw AUX status and telemetry data

(Notes) This is usually within 20 msec of the query, so the lag is small. But the update interval for AUX telemetry is typically 200 msec, so the data can be stuff acquired maximum 0.2 seconds ago.

ACMD send a raw AUX control remote command, arg: <acmd>

(Usage) acmd <acmd>

<acmd>: AUX raw commands, consist of <subsys> <auxcmd>

(Details) AUX raw commands are defined in the "KMTNet Auxiliary control software – Remote commands definition" document. <acmd> doesn't include <Telescope ID>, <System> and <Sequence Num> defined in the protocol.

(Notes) This command returns the completion acknowledgement with 'DONE' message, if the raw command was just sent without error. The final result for the operation should be checked with the status and telemetry data by ASTAT or AUXSTATUS.

Refer to section (4) for more details of the raw AUX status and telemetry data

FILTER change filters to designated filter number with an argument

(Usage) filter <fnum>

<fnum>: filter number to set, 0 ~ 4 (0: no filter)

(Details)

(Notes) The final result of this operation should be checked using FSASTAT, ASTAT or AUXSTATUS.

FSASTAT query & return the raw AUX Filter/Shutter status

(Usage) fsastat

(Details) lightweight ASTAT for only Filter/Shutter status

The last-updated Filter/Shutter status including diagnostic data are returned, or just the time/date info and ARC mode setting are returned if the AUX link is down. This command returns a lightweight status string in simple, non-IMPv2-compliant format for simple reading/parsing.

The format is as follows, depending on the AUX link state:

AUX_UP:

UP FOPSTAT FNUMBER SOPSTAT SSTATUS

AUX_DOWN:

DOWN

Refer to section (4) for more details of the raw Filter/Shutter status

(Notes) This is usually within 20 msec of the query, so the lag is small. But the update interval for AUX telemetry is typically 200 msec, so the data can be stuff acquired 0.2 sec ago(maximum).

DFOCUS adjust the focus position of PFI as delta position (offset)

(Usage) dfocus <dfoc>

<dfoc>: delta focus = dest.focus - curr.focus, $\pm x.xxx$ (mm)

(Details) The focus position is calculated for center of the prime focus instrument (on optical axis). It is same as average of 3 actuator's absolute position.

(Notes) The final result of this operation should be checked using FTTSTAT, ASTAT or AUXSTATUS.

DTILT adjust the tip-tilt angle of PFI as delta tilt

(Usage) dtilt <dtns> <dtew>

<dtns>: delta tilt for N-S, $\pm x.x$ (arcsec)

<dtew>: delta tilt for E-W, $\pm x.x$ (arcsec)

Delta tilt = dest.tilt angle - curr.tilt angle

(Details) allowed only if EXEC from remote hosts, the tip-tilt angle is positive when N/E goes up and S/W goes down. For this command, the actuator number setting for the orientation is defined correctly in the runtime configuration.

(Notes) The final result of this operation should be checked using FTTSTAT, ASTAT or AUXSTATUS.

FTTGOTO goto the absolute position of the Focus/Tip-tilt

(Usage) fttgoto <foc> (<tns> <tew>)

<foc>: absolute focus position, $\pm x.xxx$ (mm)

<tns>: abs tilt angle for N-S, $\pm x.x$ (arcsec)

<tew>: abs tilt angle for E-W, $\pm x.x$ (arcsec)

<tns> & <tew> arguments are optional and allowed only EXEC.

(Details) This command is allowed only if EXEC from remote hosts, if including the arguments for tip-tilt.

(Notes) The final result of this operation should be checked using FTTSTAT, ASTAT or AUXSTATUS.

FTTSTAT query & return raw AUX Focuser/Tip-tilt status

(Usage) fttstat

(Details) lightweight ASTAT for only Focus/Tip-tilt status

The last-updated Focus/Tip-tilt status including diagnostic data are returned, or just the time/date info and ARC mode setting are returned if the AUX link is down. This command returns a lightweight status string in simple, non-IMPv2-compliant format for simple reading/parsing.

The format is as follows, depending on the AUX link state:

AUX_UP:

```
UP FASTAT FOCUS TILTNS TILTEW SOUTHLIM EASTLIM WESTLIM
SOUTHPOS EASTPOS
```

AUX_DOWN:

```
DOWN
```

Refer to section (4) for more details of the raw Focus/Tip-tilt status

(Notes) This is usually within 20 msec of the query, so the lag is small. But the update interval for AUX telemetry is typically 200 msec, so the data can be stuff acquired 0.2 sec ago(maximum).

(4) Status and Telemetry data strings

Link STATUS

```
UP      the link is active and the telemetry data being updated
IDLE    the link has been idle for longer than the allowed time
DOWN    the link is disabled or TCP/IP connection is disconnected
```

TCSSTATUS

TCS_UP:

```
DATE-OBS=YYYY-MM-DD TIME-OBS=HH:MM:SS.SSS TIMESYS=UTC
DATE-UP=YYYY-MM-DD TIME-UP=HH:MM:SS.SSS
RA=HH:MM:SS.SSS DEC=±DD:MM:SS.SS EQUINOX=YYYY.YYY
HA=±HH:MM:SS ST=HH:MM:SS SECZ=X.XX ALT=XX.X AZ=±XXX.X
TCSLINK=Up ARCMODE=Enabled/Disabled
TELMOVE=RA/Dec/RA+Dec/Idle (TCSLIMIT=RA/TCSLIMIT=Dec/TCSLIMIT=Horizon)
TCSDRIVE=Enabled/Disabled COMNUM=X EXECODE= /e/E/3
```

TCS_IDLE:

DATE-OBS=YYYY:MM:DD TIME-OBS=HH:MM:SS.SSS TIMESYS=UTC
TCSLINK=Idle ARCMODE=Enabled/Disabled

TCS_DOWN:

DATE-OBS=YYYY:MM:DD TIME-OBS=HH:MM:SS.SSS TIMESYS=UTC
TCSLINK=Down ARCMODE=Enabled/Disabled

(Descriptions)

DATE-OBS: current date recorded when this function is called
TIME-OBS: current time recorded when this function is called
DATE-UP : updated date recorded when the telemetry packet was received
TIME-UP : updated time recorded when the telemetry packet was received
EXECODE : ' ' / 'e' / 'E' / '3',
 if raw command was executed successfully, changed to e/E
 if raw command was unrecognized, changed to 3

TSTAT

TCS_UP:

UP DATE-OBS TIME-OBS TIMESYS DATE-UP TIME-UP ARCMODE
RA DEC EQUINOX HA ST SECZ ALT AZ MOVESTATUS EXECODE

TCS_IDLE:

IDLE DATE-OBS TIME-OBS TIMESYS ARCMODE

TCS_DOWN:

DOWN DATE-OBS TIME-OBS TIMESYS ARCMODE

(Descriptions)

DATE-OBS: current date recorded when this function is called
TIME-OBS: current time recorded when this function is called
DATE-UP : updated date recorded when the telemetry packet was received
TIME-UP : updated time recorded when the telemetry packet was received
TIMESYS : UTC
MOVESTATUS : 0:no / 1:RA moving / 2:Dec moving / 3: Both moving
EXECODE : ' ' / 'e' / 'E' / '3',
 if raw command was executed successfully, changed to e/E
 if raw command was unrecognized, changed to 3
ARCMODE : AutoRecovery mode flag, ARC.ENABLED / ARC.DISABLED

The other value's format and unit is same as TCSSTATUS string.

AUXSTATUS

AUX_UP & all subsystems are connected to the device:

DATE-OBS=YYYY-MM-DD TIME-OBS=HH:MM:SS.SSS TIMESYS=UTC
DATE-UP=YYYY-MM-DD TIME-UP=HH:MM:SS.SSS
AUXLINK=Up ARCMODE=Enabled/Disabled
FILTERSHUTTER_STATUS=STANDBY/RUNNING/ERROR
FILTER_OPSTAT=STANDBY/RUNNING/ERROR
FILTERNUM=0/1/2/3/4/5/-1 (0: no / 5: 2 more / -1: unknown)
SHUTTER_OPSTAT=STANDBY/OPENING/OPENED/CLOSING/RELOADING/ERROR
SHUTTER=OPEN/CLOSED

FOCUSACTUATOR_STATUS=STANDBY/RUNNING/ERROR
 FA_FOCUS= $\pm$ X.XXX (mm)
 FA_TILT_NS= $\pm$ X.X (arcsec)
 FA_TILT_EW= $\pm$ X.X (arcsec)
 FA_SOUTH_LIMIT=0/1/2/3 (0: all off / 1: outer / 2: inner / 3: both)
 FA_EAST_LIMIT=0/1/2/3 (0: all off / 1: outer / 2: inner / 3: both)
 FA_WEST_LIMIT=0/1/2/3 (0: all off / 1: outer / 2: inner / 3: both)
 FA_SOUTH_POS= $\pm$ X.XXX (mm)
 FA_EAST_POS= $\pm$ X.XXX (mm)
 FA_WEST_POS= $\pm$ X.XXX (mm)

DOMESHUT_STATUS=STANDBY/RUNNING/ERROR

DS_UPSHUT=OPEN/MID/CLOSED
 DS_LOSHUT=OPEN/MID/CLOSED
 DS_SAFSW=ACTIVE/INACTIVE
 DS_AUTOSYNC=ENABLED/DISABLED
 DS_SHUT_ALT=XX.X (deg)
 DS_TELE_ALT=XX.X (deg)

MIRRORCOVER_STATUS=STANDBY/RUNNING/ERROR

MC_POS=XXX (%)

CHILLER_STATUS=STANDBY/ERROR

CH_COOLING=ON/OFF
 CH_SETPOINT=XX.X (deg C)
 CH_PROCTEMP=XX.X (deg C)

ENVIRONMENT_STATUS=STANDBY/ERROR

EN_FAN=ON/OFF
 EN_S1=XX.X (deg C/RH %)
 EN_S2=XX.X (deg C/RH %)
 EN_S3=XX.X (deg C/RH %)
 EN_S4=XX.X (deg C/RH %)
 EN_S5=XX.X (deg C/RH %)
 EN_S6=XX.X (deg C/RH %)
 EN_S7=XX.X (deg C/RH %)

AUX_UP & all subsystems are disconnected from the device:

DATE-OBS=YYYY-MM-DD TIME-OBS=HH:MM:SS.SSS TIMESYS=UTC
 DATE-UP=YYYY-MM-DD TIME-UP=HH:MM:SS.SSS
 AUXLINK=Up ARCMODE=Enabled/Disabled
 FILTERSHUTTER_STATUS=NC
 FOCUSACTUATOR_STATUS=NC
 DOMESHUT_STATUS=NC
 MIRRORCOVER_STATUS=NC
 CHILLER_STATUS=NC
 ENVIRONMENT_STATUS=NC

AUX_DOWN:

DATE-OBS=YYYY-MM-DD TIME-OBS=HH:MM:SS.SSS TIMESYS=UTC
 AUXLINK=Down ARCMODE=Enabled/Disabled

(Descriptions)

DATE-OBS: current date recorded when this function is called
 TIME-OBS: current time recorded when this function is called
 DATE-UP : updated date recorded when the telemetry packet was received

TIME-UP : updated time recorded when the telemetry packet was received

<abbreviations>

FS: Filter/Shutter box
 FA: Focus Actuator
 DS: Dome Shutter
 MC: Mirror Cover
 CH: Chiller for mirror cooling
 EN: Environment monitor

ASTAT

AUX_UP:

UP DATE-OBS TIME-OBS TIMESYS DATE-UP TIME-UP ARCMODE
 FS: FSSTAT FOPSTAT FNUMBER SOPSTAT SSTATUS
 FA: FASTAT FOCUS TILTNS TILTEW SOUTHLIM EASTLIM WESTLIM
 SOUTHPOS EASTPOS WESTPOS
 DS: DSSTAT UPSHUT LOSHUT SAFSW AUTOSYNC SHUTALT TELEALT
 MC: MCSTAT MCPOS
 CH: CHSTAT COOLING SETPOINT PROCTEMP
 EN: ENSTAT FAN S1 S2 S3 S4 S5 S6 S7

AUX_DOWN:

DOWN DATE-OBS TIME-OBS TIMESYS ARCMODE

(Descriptions)

DATE-OBS : current date recorded when this function is called
 TIME-OBS : current time recorded when this function is called
 DATE-UP : updated date recorded when the telemetry packet was received
 TIME-UP : updated time recorded when the telemetry packet was received
 TIMESYS : UTC
 FS: : Filter/Shutter data start
 FSSTAT : FS subsystem operation status (NC/STANDBY/RUNNING/ERROR)
 FOPSTAT : filter operation status (NC/STANDBY/RUNNING/ERROR)
 FNUMBER : current filter number
 (0: no / 1~4: filter #1~4 / 5: 2 more / -1: unknown)
 SOPSTAT : shutter operation status
 (NC/STANDBY/OPENING/OPENED/CLOSING/RELOADING/ERROR)
 SSTATUS : shutter status (OPEN/CLOSED/UNKNOWN)
 FA: : Focuser Actuator data start
 FASTAT : FA subsystem operation status (NC/STANDBY/RUNNING/ERROR)
 FOCUS : focus position at the center of PFI, average of 3 actuator pos
 TILTNS : North-South PFI tilt angle in arcsec,
 positive when N is higher than S
 TILTEW : East-West PFI tilt angle in arcsec,
 positive when E is higher than W
 SOUTHLIM : limit status of south actuator (no:0/outer:1/inner:2/both:3)
 EASTLIM : limit status of east actuator
 WESTLIM : limit status of west actuator
 SOUTHPOS : position of south actuator in mm
 EASTPOS : position of east actuator in mm

WESTPOS : position of west actuator in mm
 DS: : Dome Shutter data start
 DSSTAT : DS subsystem operation status (NC/STANDBY/RUNNING/ERROR)
 UPSHUT : upper dome shutter status (OPEN/MID/CLOSED)
 LOSHUT : lower dome shutter status (OPEN/MID/CLOSED)
 SAFSW : safety interlock switch status (ACTIVE/INACTIVE)
 AUTOSYNC : dome shutter-telescope Auto-sync mode (ENABLED/DISABLED)
 SHUTALT : upper dome shutter altitude in deg
 TELEALT : telescope altitude that AUX read from Telcom in deg
 MC: : Mirror Cover data start
 MCSTAT : MC subsystem operation status (NC/STANDBY/RUNNING/ERROR)
 MCPOS : mirror cover position (0~100)
 CH: : Chiller for mirror cooling data start
 CHSTAT : CH subsystem operation status (NC/STANDBY/ERROR)
 COOLING : chiller cooling switch status (ON/OFF)
 SETPOINT : chiller set point temperature, in deg C
 PROCTEMP : chiller processed temperature, in deg C
 EN: : Environmental system data start
 ENSTAT : EN subsystem operation status (NC/STANDBY/ERROR)
 FAN : mirror cooling fan relay status (ON/OFF)
 S1 ~ S7 : Environmental sensor #1 ~ #7 data, in deg C or RH %
 ARCMODE : AutoRecovery mode flag (ARC.ENABLED/ARC.DISABLED)

The value format and unit is same as AUXSTATUS string.

FSASTAT

AUX_UP:
 UP FOPSTAT FNUMBER SOPSTAT SSTATUS
 AUX_DOWN:
 DOWN

(Descriptions)

FOPSTAT : filter operation status (NC/STANDBY/RUNNING/ERROR)
 FNUMBER : current filter number
 (0: no / 1~4: filter #1~4 / 5: 2 more / -1: unknown)
 SOPSTAT : shutter operation status
 (NC/STANDBY/OPENING/OPENED/CLOSING/RELOADING/ERROR)
 SSTATUS : shutter status (OPEN/CLOSED/UNKNOWN)

The value format and unit is same as AUXSTATUS string.

FTTSTAT

AUX_UP:
 UP FASTAT FOCUS TILTNS TILTEW SOUTHLIM EASTLIM WESTLIM
 SOUTHPOS EASTPOS WESTPOS
 AUX_DOWN:
 DOWN

(Descriptions)

FASTAT : focuser operation status (NC/STANDBY/RUNNING/ERROR)
 FOCUS : focus position at the center of PFI, average of 3 actuator pos
 TILTNS : North-South PFI tilt angle in arcsec,
 positive when N is higher than S
 TILTEW : East-West PFI tilt angle in arcsec,
 positive when E is higher than W
 SOUTHLIM : limit status of south actuator (no:0/outer:1/inner:2/both:3)
 EASTLIM : limit status of east actuator
 WESTLIM : limit status of west actuator
 SOUTHPOS : position of south actuator in mm
 EASTPOS : position of east actuator in mm
 WESTPOS : position of west actuator in mm

The value format and unit is same as AUXSTATUS string.

6. Operation

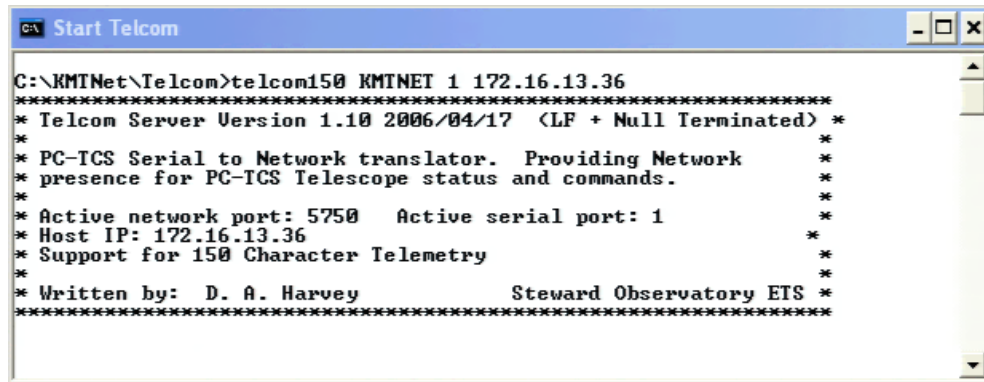
6.1. Notes for TCS operation and software test

- In the camera shutter operation, the camera control software should monitor and check the Filter/Shutter status because it must be reflected in the operation. The camera shutter is not controlled by TCS Agent. It is controlled by TTL shutter input via hardware interface. The ICS controls the HE box utility for the shutter OPEN/CLOSE operation by setting the shutter input to HIGH/LOW. In these shutter operation, the OPEN (shutter input: LOW→HIGH) must be commanded when the shutter operation status is STANDBY. And also the CLOSE (shutter input: HIGH→LOW) must be commanded when the shutter operation status is OPENING or OPENED, to prevent an operational error on the Filter/Shutter device.
- TCS Agent sets the Input Epoch of PC-TCS to J2000 automatically at TCS link initialization. But when PC-TCS is not running or not in the normal operation, the automatic Input Epoch setting can be not successful and TCS Agent displays a message for the Input Epoch setting failure. If the Input Epoch setting is not initialized successfully, the Input Epoch must be set to J2000 manually on PC-TCS before commanding 'TGOTO' or 'TOFFSET' to slew the telescope.
- The user must check the UT date will not change soon before using 'TSYNC' keyboard command. If the time is pass on 24:00 in progress, the date will be not correct since time-sync command and date-sync command are separated and there is some delay.

- TSTOP and TDI commands can be used in the software operational test that is performed without the telescope using the original version of PC-TCS that does not have DEMO function.
- The AUX control simulator program provides some raw commands for the software operational test performed without the subsystems. The AUX simulation commands 'CSHUT HIGH/LOW' is to simulate the TTL input for camera shutter operation. The AUX simulation 'STATERR <SUBSYS>' and 'CLEARERR' are to simulate the operational error of subsystems. The original AUX control remote server does not provide these AUX simulation commands.

6.2. Telcom and AUX control program

Telcom is a Win32 console application. It is executed by typing "telcom150 KMTNET <COM port number of AUX computer> <AUX host IP>" on the command prompt of the Window system. The "starttelcom.bat" in the 'telcom' directory can be used for a template of the batch file to lunch a Telcom program. The following panel shows the Telcom application executed by the "starttelcom.bat" batch.



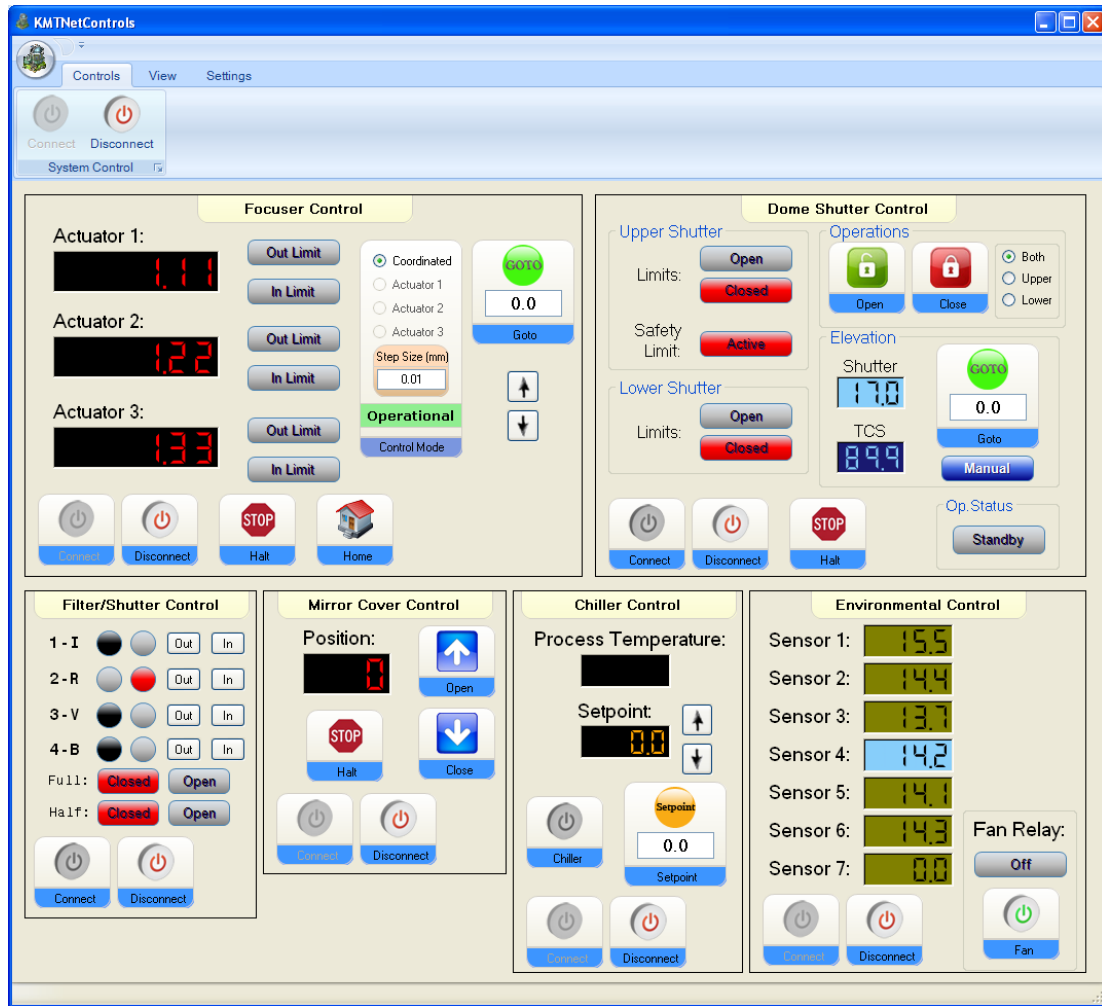
```

C:\KMTNet\Telcom>telcom150 KMTNET 1 172.16.13.36
*****
* Telcom Server Version 1.10 2006/04/17 (LF + Null Terminated) *
*
* PC-TCS Serial to Network translator. Providing Network
* presence for PC-TCS Telescope status and commands.
*
* Active network port: 5750 Active serial port: 1
* Host IP: 172.16.13.36
* Support for 150 Character Telemetry
*
* Written by: D. A. Harvey Steward Observatory ETS
*****

```

<PC-TCS Telcom, executed on the Win-command prompt>

The following panel is the GUI of AUX control software. The AUX server is included in this program. The IP address and port number of TCP server for remote interaction can be configured in the settings tab. The shortcut 'KMTNetControls' can be used for launching the program on the desktop screen of the AUX host computer.



<AUX control software – GUI control panel>

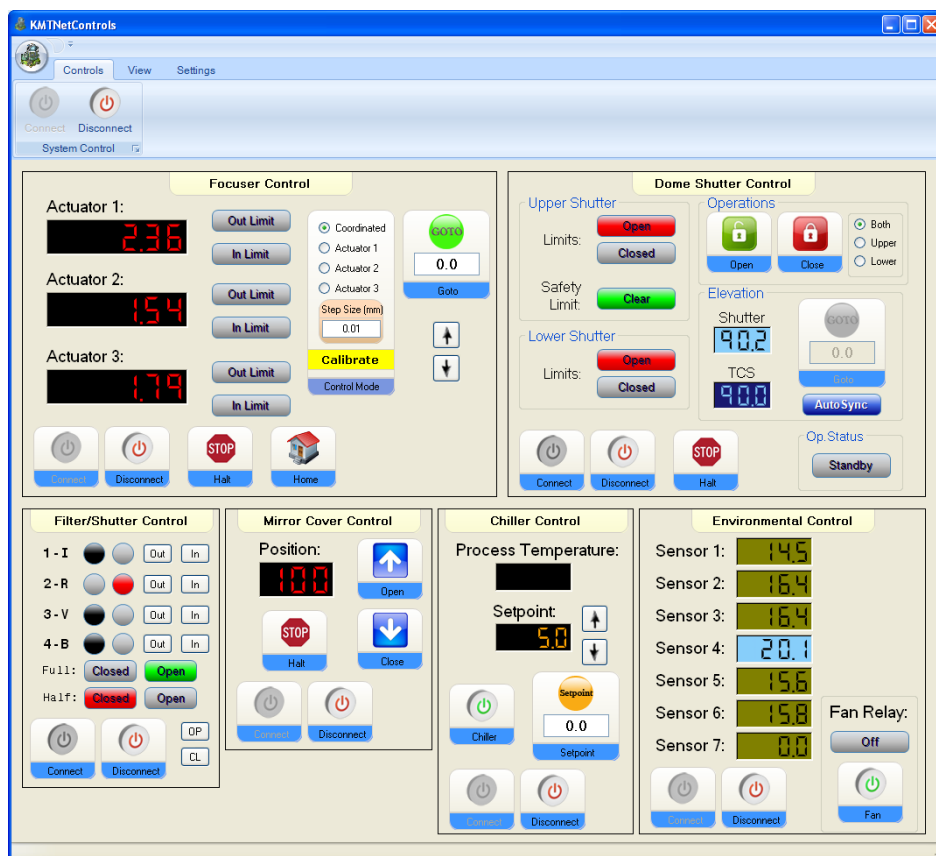
6.3. Simulation and Operational Test

Software operational test for the KMTNet TCS and TCS Agent can be performed through the TCS simulation configuration without the telescope and AUX subsystems. The original versions of PC-TCS and Telcom can be used for the operational test, but the original AUX control program is not available because there are no subsystems and the connection with devices also cannot be performed. So the demo version of AUX control software is used in the test.

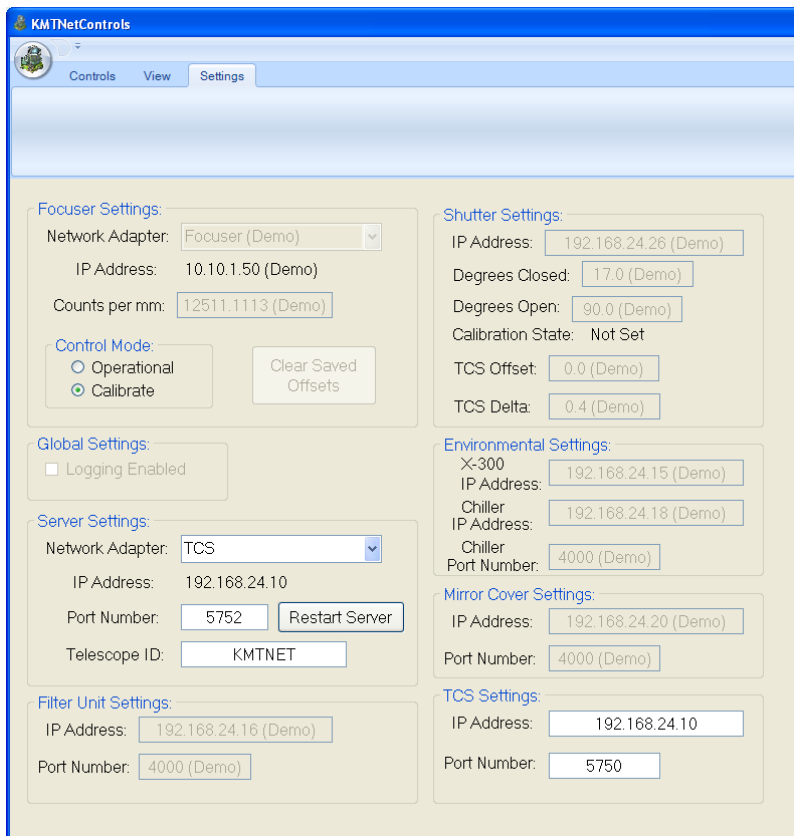
The PC-TCS demo version is more useful than the original version for the telescope control simulation. If the demo version is not available, the original version can also be used for the operational test. In the case of original version, when PC-TCS is commanded to move the telescope, the moving status is changed to 1 on RA axis, Dec axis or both axes. Then TCS Agent understands the telescope is moving to commanded position. The commanded motion is never completed and the moving status is also not changed to 0 because there is no the telescope. In this situation, the TDI command is useful to make the current telescope position to the

commanded position on PC-TCS as if the commanded motion is completed. The TSTOP command can be also used to cancel the commanded motion.

The AUX demo program for the AUX control interface test is working on Windows XP or 7. On the system, Microsoft .NET frame work 3.5 or later version must be installed to run the original version or the demo version of AUX control software. The GUI control panel is same as the original version and simulated operational statuses of all subsystems are displayed on the panel in real time. The interface and protocol for remote interaction is also totally same as the original AUX control software. At the first try for running the program, TCP Server Settings and TCS Settings is configured on the Settings panel. The IP address can be set with the Ethernet adaptor setting or internal local host IP address (127.0.0.1) of the AUX simulation computer (Windows XP or 7). The port number should be set to 5752 in Server Settings and 5750 in TCS Settings. The telescope ID should be set to 'KMTNET'. The other settings are disabled in the demo version. The following figures show the GUI control panel and setting panel of AUX control simulator program.



<AUX control Demo program – Controls panel>



<AUX control Demo program – Settings panel>

The AUX control demo program provides some raw commands for the software operational test and hardware simulation. The 'CSHUT HIGH/LOW' command is to simulate the TTL input for camera shutter operation. The 'STATERR <SUBSYS>' and 'CLEARERR' commands are to simulate the operational error of subsystems. In the original version, the remote TCP server does not provide these simulation commands.

<AUX simulation commands list>

Subsystem	Command	Parameters	Response	Notes	Idx
SIMUL	CSHUT	<action>	<ACK>	<action>: HIGH/LOW	7-1
	STATERR	<subsys>	<ACK>	<subsys>: AL/FS/FA/DS/MC/CH/EN	7-2
	CLEARERR	-	<ACK>	Clear all subsystems error	7-3

<Examples for AUX simulation commands>

```
IS>TC fsastat
TC>IS DONE: STANDBY 1 STANDBY CLOSED

IS>TC acmd simul cshut high
TC>IS DONE: OK

IS>TC fsastat
```

```

TC>IS DONE: STANDBY 1 OPENING OPEN

IS>TC fsastat
TC>IS DONE: STANDBY 1 OPENED OPEN

IS>TC acmd simul cshut low
TC>IS DONE: OK

IS>TC fsastat
TC>IS DONE: STANDBY 1 CLOSING OPEN

IS>TC fsastat
TC>IS DONE: STANDBY 1 RELOADING CLOSED

IS>TC fsastat
TC>IS DONE: STANDBY 1 STANDBY CLOSED

IS>TC fttstat
TC>IS DONE: STANDBY +0.0 +0.0  0 0 0  +0.000 +0.000 +0.000

IS>TC acmd simul staterr fa
TC>IS DONE: OK

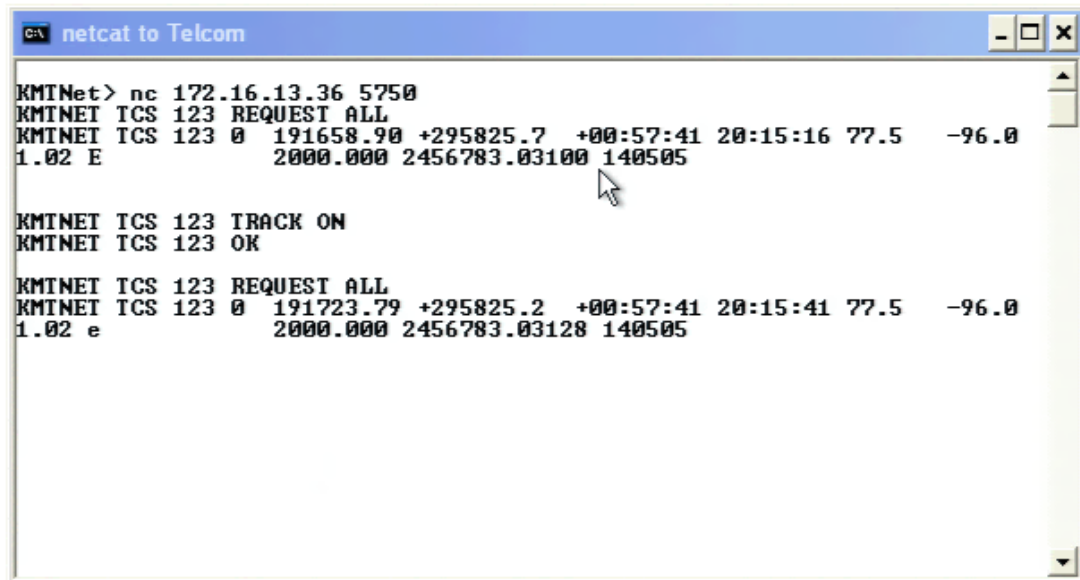
IS>TC fttstat
TC>IS DONE: ERROR +0.0 +0.0  0 0 0  +0.000 +0.000 +0.000

IS>TC acmd simul clearerr
TC>IS DONE: OK

IS>TC fttstat
TC>IS DONE: STANDBY +0.0 +0.0  0 0 0  +0.000 +0.000 +0.000

```

Netcat can be used for basic communication check in the various cases. Netcat, a GNU open-source program, is a useful utility for the network TCP/UDP communication test. The installation files and open source can be downloaded with manuals on the Web.



```

C:\ netcat to Telcom

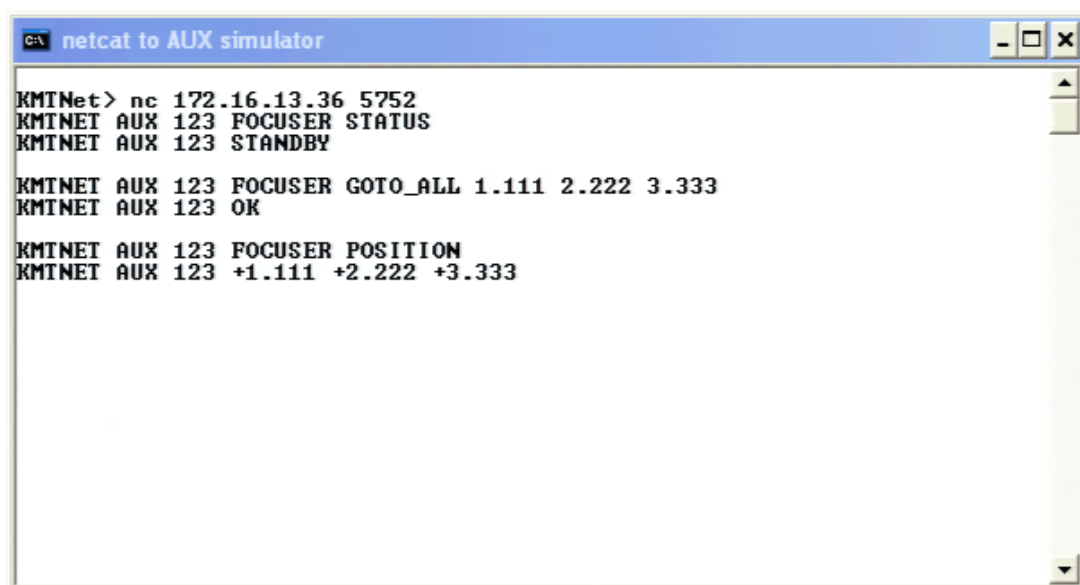
KMTNet> nc 172.16.13.36 5750
KMTNET TCS 123 REQUEST ALL
KMTNET TCS 123 0 191658.90 +295825.7 +00:57:41 20:15:16 77.5 -96.0
1.02 E 2000.000 2456783.03100 140505

KMTNET TCS 123 TRACK ON
KMTNET TCS 123 OK

KMTNET TCS 123 REQUEST ALL
KMTNET TCS 123 0 191723.79 +295825.2 +00:57:41 20:15:41 77.5 -96.0
1.02 e 2000.000 2456783.03128 140505

```

<Example for communication with Telcom using Netcat on Windows>



```

C:\ netcat to AUX simulator

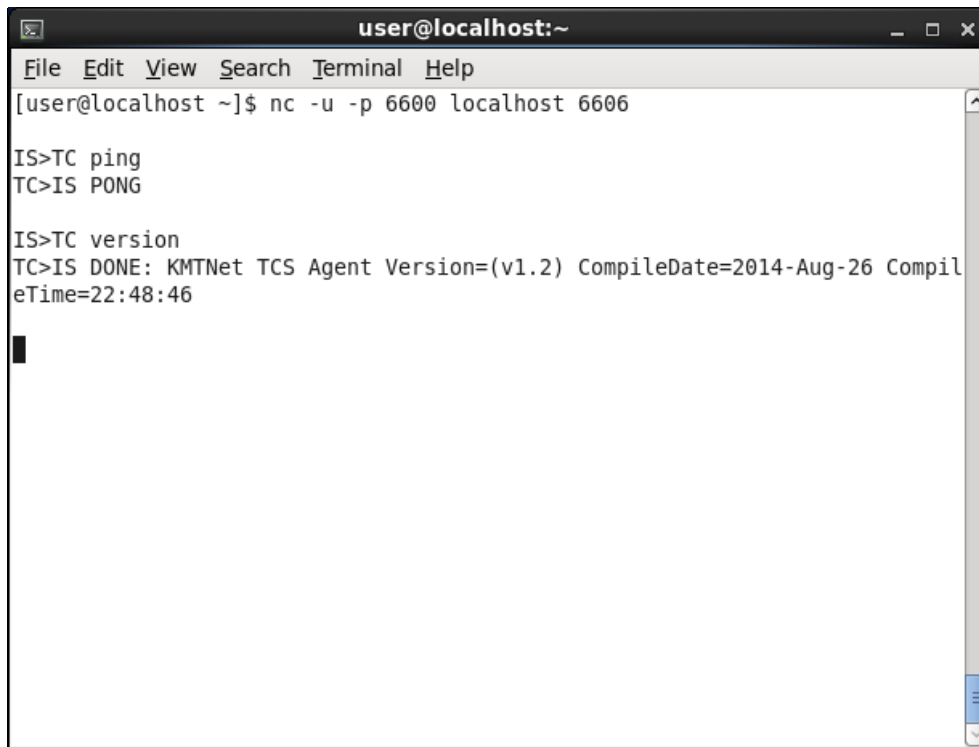
KMTNet> nc 172.16.13.36 5752
KMTNET AUX 123 FOCUSER STATUS
KMTNET AUX 123 STANDBY

KMTNET AUX 123 FOCUSER GOTO_ALL 1.111 2.222 3.333
KMTNET AUX 123 OK

KMTNET AUX 123 FOCUSER POSITION
KMTNET AUX 123 +1.111 +2.222 +3.333

```

<Example for communication with AUX server using Netcat on Windows>



```
user@localhost:~  
File Edit View Search Terminal Help  
[user@localhost ~]$ nc -u -p 6600 localhost 6606  
  
IS>TC ping  
TC>IS PONG  
  
IS>TC version  
TC>IS DONE: KMTNet TCS Agent Version=(v1.2) CompileDate=2014-Aug-26 CompileTime=22:48:46  
█
```

<Example for communication with AUX server using Netcat on Linux>

Rev1, 2014-05-12 S. Cha

Rev2, 2014-08-27 S. Cha

Rev3, 2014-09-07 S. Cha

Rev4, 2020-11-10 S. Cha

KMTNet operation team in KASI